

Exam 1 review

ECON-4210

Economics of Climate Change

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Review...

- Benefit-cost analysis, continued
- Exam 1: format and coverage
- Problem set 1 questions?
- Practice problems

Exam 1

Monday, September 21, in class

- 20 multiple choice questions, 1 point each.
- 2 short answer questions, 10 points each.
 - Show your work. Partial credit for a correct method with an arithmetic error.
- One-page cheat sheet. No other references, books, or technology.
- You may use a calculator.

What the exam covers

Classes 1–6 and Problem Set 1

- 01 – Overview of the course
- 02 – What is climate change? Climate science 101
- 03 – How are the climate and the economy linked?
- 04 – Why is climate change an economic problem? (I)
- 05 – Why is climate change an economic problem? (II)
- 06 – What are the costs and benefits of climate policy?
 - ▶ Readings are fair game if we've covered them in class.

Practice problem 1

Externalities in coal-fired electricity

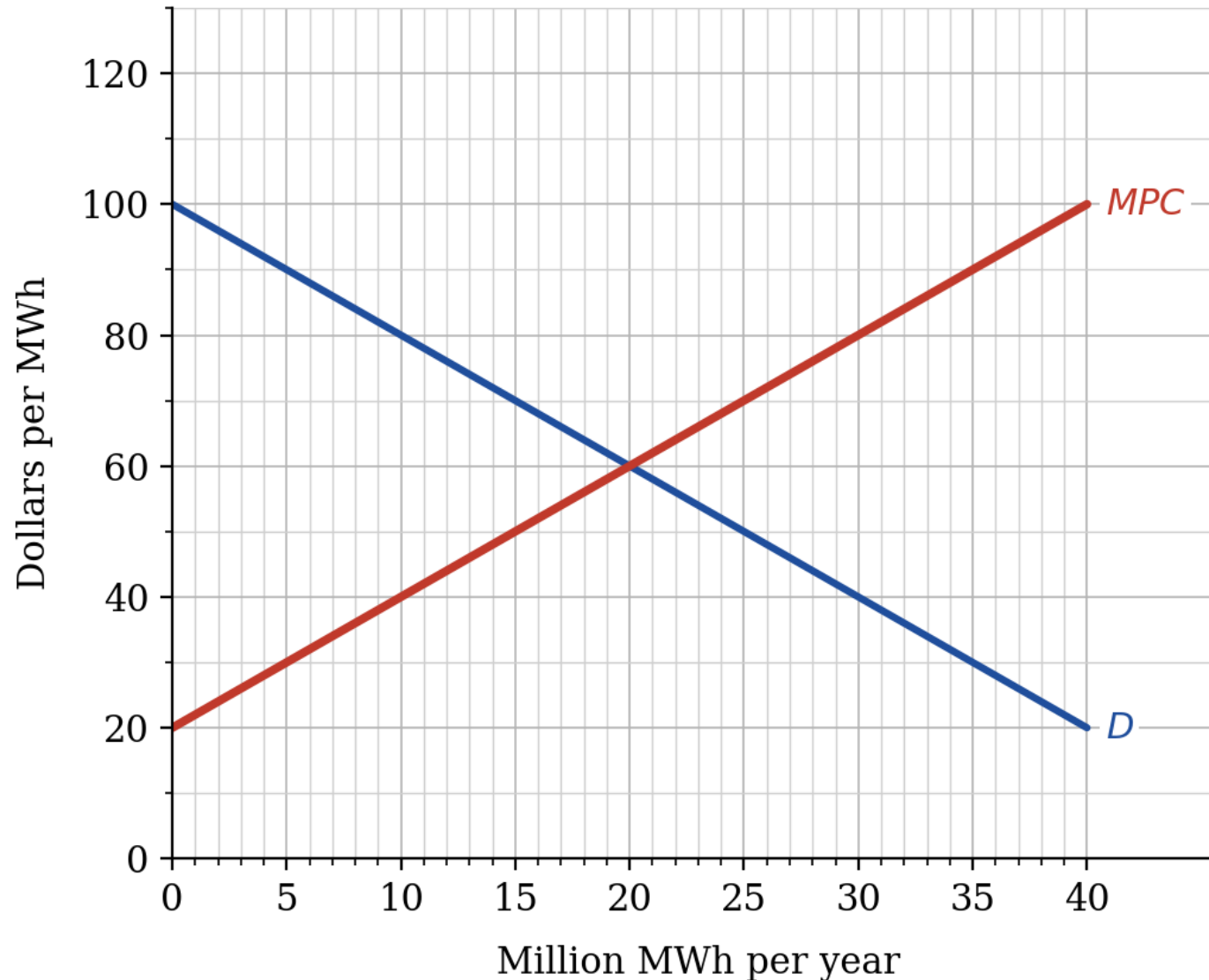
- Annual demand and marginal private cost for electricity from a coal-fired generator. Quantity is million MWh per year, dollar figures are per MWh.
- $D = 100 - 2Q$ $MPC = 20 + 2Q$
- Burning coal releases greenhouse gases with marginal external damages of \$20 per MWh.
 - ▶ (a) Equilibrium price and quantity in this market.
 - ▶ (b) Draw and label MSC. Socially optimal quantity, and the price that generates it.
 - ▶ (c) Shade and label the deadweight loss. What is its dollar value?
 - ▶ (d) What's the per-unit tax that would produce the socially optimal Q, and the price consumers pay?

Practice problem 1

$$D = 100 - 2Q$$

$$MPC = 20 + 2Q$$

External damage: \$20 per MWh



- (a) Equilibrium price and quantity.
- (b) Draw MSC. Find Q^* and the price.
- (c) Shade/quantify the deadweight loss.
- (d) What tax gets us to Q^* ?

Practice problem 2

Public goods

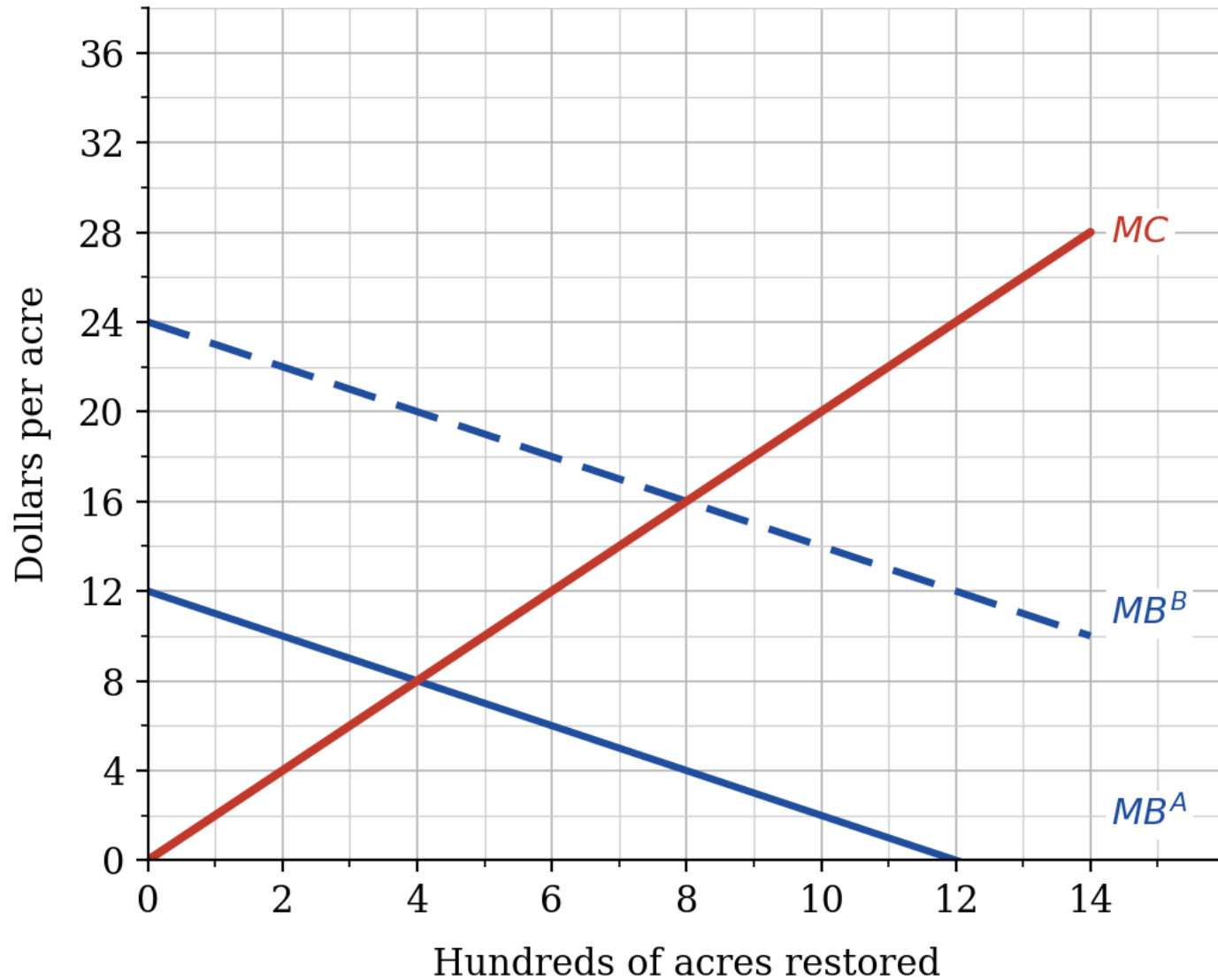
- Towns A and B sit downstream of the same wetland. Restoring acreage filters runoff for both at once, and neither town can be kept from the improvement.
- $MB^A = 12 - Q$ $MB^B = 24 - Q$ $MC = 2Q$
- Acreage is in hundreds of acres. Dollar figures are per acre.
 - ▶ (a) Acting alone, how much would Town A restore? Town B?
 - ▶ (b) Draw and label SMB.
 - ▶ (c) What is the efficient level of restoration?
 - ▶ (d) Town B has already restored its amount from (a). Does Town A add any?

Practice problem 2

$$MB^A = 12 - Q$$

$$MB^B = 24 - Q$$

$$MC = 2Q$$



(a) How much does each town restore alone?

(b) Draw SMB.

(c) The efficient level. Verify it.

(d) B restores its amount first. Does A add any?

Practice problem 3

Present value and the NPV rule

- A county can install a gas capture system at its landfill for \$18 million today. Methane emissions rise as the landfill fills, so avoided climate damages grow:
- $D = 2 + Y$, where D is avoided damages in \$ millions and Y is the year.
- The system operates for five years. Use a discount rate of 5%.
 - ▶ (a) Avoided damages in each of the five years.
 - ▶ (b) Discount each year at 5%.
 - ▶ (c) Present value of avoided damages over the five years.
 - ▶ (d) Should the county install the system? Explain.

Practice problem 3

$D = 2 + Y$, in \$ millions. Discount rate 5%. The system costs \$18 million today.

Year (Y)	Avoided damages	Discounted
1		
2		
3		
4		
5		